Draft version April 11, 2025
Typeset using LaTeX **twocolumn** style in AASTeX631

Simulating the signals from SGWB model produced during inflation

Jiayi Xu and Supervisors: Renée Hložek, Simran Nerval

ABSTRACT

Inflation is one of the most important stages of the early universe, as it lays the foundation for the evolution of the universe. We do not have definitive observational evidence for any particular inflationary model. One way to study inflation is to compute the observational signatures from a given inflationary model. We perform a lattice simulation and obtain the predicted stochastic gravitational wave background (SGWB) signals from specific models of inflation. We focus on simulating the signatures from preheating/reheating for the α-attractor T-model potential. From the result we obtained, we observed the formation of "oscillons" which is a special structure that can store energy for a very long time at a specific location. The frequency range of the simulated signals are bounded between 10^7 to 10^{10} Hz, and the energy density range is about 10^{-21} to 10^{-25} for non-oscillons signals, and 10^{-9} to 10^{-13} for oscillons signals. The tensor-to-scalar value at which we observe oscillon formation is $r \lesssim 5 \times 10^{-6}$. Below this value will favor the self-interaction of the field.

1. INTRODUCTION

In 2015, astronomers were thrilled by a groundbreaking discovery: the direct detection of gravitational waves by the Laser Interferometer Gravitational-Wave Observatory (LIGO) (Abbott et al. 2016). This began a new era of multi-messenger astronomy because gravitational waves contain lots of information about the dynamics of different astronomical events and objects. Gravitational waves (GW) have weak interaction with matter, which allows it to travel vast distances. As a result, we are able to study the evolutionary process of the early universe from the unique signals it emitted. Unlike lights, CMB only provides information after electrons been captured and the universe become transparent, but the stochastic gravitational wave background (SGWB) contains information even earlier than recombination (Bailes et al. 2021), which could be potentially directly probe in the future.

First proposed in the 1980s (Guth), inflation is a key component of the big bang model and solves the horizon, flatness and monopole problems (Guth 1981). This makes inflation a fundamental process of the evolution of the early universe, and became a foundation of the evolution process of a de Sitter universe (R. Liddle 1999). One suggested piece of evidence for inflation is the SGWB (García-Bellido et al. 2017). The motivation of this project is to compute and predict the spectrum of the SGWB from particular models of inflation. Thus, we will be able to better constrain the inflationary models and support the theory of inflation by comparing simulation result with SGWB constrains and current detectable range.

The following sections will introduce how GW were generated during inflation (Section 2.1), both slow-roll inflation and preheating/reheating periods (Section 2.2), and the formation of oscillons (section 2.3). Section 3 introduced the tool used to compute the simulation, as well as the procedure to approach the goal. The GW signals obtained will be show in Section 4, and lastly, Section 5 concludes this project and gives some future implications.

2. BACKGROUND

2.1. *Generation of gravitational waves*

GW are the transverse waves that travel at speed of light due to the distortions of the curved spacetime. They manifest as a time-dependence strain on spacetime. They carry unique messages about the dynamics of massive objects (Bailes et al. 2021). SGWB is a collection of continuous, weak GW emitted from different sources. It contains the information of the overall structure of the early universe covering a wide range of frequencies (Guzzetti et al. 2016). That being said, the GW generated by inflation were also contribute to the SGWB signal, and for preheating/reheating period, the frequency range is about MHz to GHz (García-Bellido et al. 2017).

Guth's new paper in 1982 pointed out that inflation generates quantum fluctuations from Higgs field rather than completely smooth background (Guth & Pi 1982). These small fluctuations got stretched into cosmological scales which we call "perturbations", and they grow via gravitational instability into diverse structure and evolved until today. Guth stated that the exponential expansion of the universe can be represent by the Higgs field ϕ slowly rolls down the potential followed by a rapid thermalization process (Guth & Pi 1982). This will be explained in detail in Section 2.2. We call this Higgs scalar field "inflaton". The non-linear dynamic of the scalar field in de Sitter space leads to the tensor perturbation, which we can observe from the raise of energy density (Dodelson et al. 1994). These tensor perturbations also refers to the change of spacetime curvature following the General Relativistic theory (Guzzetti et al. 2016), and those corresponds to the GW we will be simulating later on.

We would like to see how much of the universe's energy density is from gravitational waves, the mathematical computation for energy density as a function of frequency is as following (Bhoonah et al. 2021):

$$\Omega_{\mathrm{gw}}(f) \equiv \frac{1}{\rho_{\mathrm{crit}}} \frac{d\rho_{\mathrm{gw}}}{d\ln(f)} \tag{1}$$

Where $\rho_{\mathrm{crit}} = 3H_0^2/(8\pi G)$ the the critical density of the universe required for a flat universe, H_0 is the Hubble constant, G is the gravitational constant, and ρ_{gw} is the (energy) density for gravitational wave that we can obtain from our simulation. We would like to plot the energy density as a function of frequency and compare with current constrains.

The strength of the signal can be quantify by the "tensor-to-scalar ratio" follow the equation,

$$r = \frac{P_{\mathrm{T}}}{P_{\mathrm{s}}} = \frac{A_{\mathrm{T}}(k/k_*)^{n_T}}{A_{\mathrm{s}}(k/k_*)^{n_s-1}} \tag{2}$$

Where r is the ratio of the power spectrum between the tensor and scalar field. k is the wavenumber, k_* is the pivot scale which we took 0.05 Mpc^{-1}, A_{T} and A_{s} are the amplitudes of the power spectra, n_T and n_s are the tensor and scalar spectral indices (Aghanim et al. 2020). Our current upper limit is $r < 0.032$ from both BICEP/Keck 2018 and Planck PR 4 data (Tristram et al. 2022). A smaller r-value corresponds to a lower energy scale, represents a fainter GW signal. Based current SGWB constrains, this energy would be bound by $\lesssim 10^{-9}$ (Abbott et al. 2019).

2.2. *Slow-roll inflation & preheating/reheating*

Inflation is a cosmological hypothesis of the evolution of the early universe. The inflation theory states that the universe expanded 50-60 e-folds in less than 10^{-32} seconds (Guth 1981; R. Liddle 1999). We quantify the expansion of the universe during inflation using what is called the e-folding number, or N, defined as $a_{\text{end of inflation}}/a_{\text{beginning of inflation}} = e^N$.

We have focused on the T-model for our simulations based on the previous study of oscillon formation in Bhoonah et al. (2021)'s paper. The potential of the T-model follows a tangential function,

$$V(\phi) = \Lambda^4 tanh^2\left(\frac{\phi}{\sqrt{6\alpha}M_{Pl}}\right) \tag{3}$$

Λ has the dimension of energy, α is a constant that determines the shape and slope of the potential, and it is related to r via $r = 12\alpha/N^2$, ϕ is the inflaton scalar field, and M_{Pl} is the reduced Planck mass (Bhoonah et al. 2021).

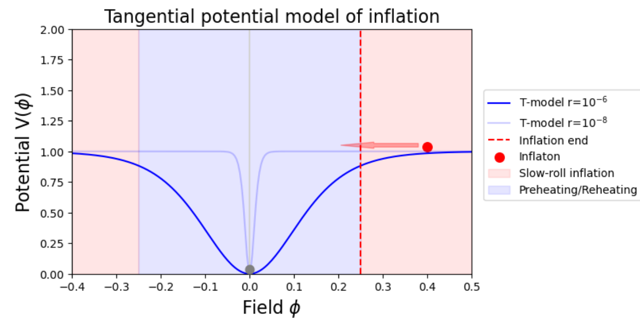

Figure 1. T-model potential followed equation 3 with for different tensor-to-scalar ratios, specifically $r = 10^{-6}$ and $r = 10^{-8}$. Slow-roll inflation (red background) and preheating/reheating (blue background) period were labeled based on how inflaton field will involve with the potential.

Figure 1 gives us a clear visualization of this symmetric-potential model as the blue curves show. The reason we choose this model is because it fit very well with the CMB data and current SGWB constrains, this will be discussed in detail in Section 4. The potential at the sections with red background has a relatively flat slope, the inflaton will slowly roll on the potential, so we call this preriod "slow-roll inflation". During this period, most energy was stored in potential to allow the universe to undergo the huge expansion process. We use the slow-roll parameters ϵ and η to quantify the slope and curvature of the potential(Bhoonah et al. 2021),

$$\epsilon = \frac{1}{2}\left(\frac{M_{Pl}V'}{V}\right)^2, \qquad \eta = \frac{M_{Pl}^2 V''}{V} \tag{4}$$

We indicate that the slow-roll inflation ends at $\epsilon \geq 1$, where the slope is no longer flat.

Right after ϵ approaches 1, the inflaton oscillates in the potential well until it eventualy settles in the minimum of the potential at the bottom with blue background. This period is called "preheating/reheating", the expansion rate of the universe became very slow. The preheating process is the initial transfer of energy from the scalar field to other particles. The following slower energy transfer is "reheating", which is the same thermalization process. Energy released from the field brings the universe back to the standard big bang theory from the short freezing, allowing particle interaction and evolution.

2.3. *Formation of oscillons*

Oscillons are massive, long-lived self-resonance configurations of a scalar field (Amin et al. 2012). Their formations are due to the self-interaction of the field, which the energy from self-interaction dominates the gravitational interaction of the field. Due to the constrains that energy must be extracted from the field and transform to particles to ensure radiation dominance of the universe at that time. Thus, those self-interaction causes large inhomogeneities (Bhoonah et al. 2021). The formation of oscillons in inflaton field density were sources of SGWB, which we will show the simulation result in this report.

From figure 1, the lighter blue potential at the back has a smaller r-value, so the shape of the potential is narrower. This promotes the formation of oscillons since a narrower potential will lead the inflaton oscillate faster in the potential well, and have a greater chance to interact with itself. But as the inflaton stops oscillating, self-interaction stops, oscillons reached their lifetime and start to decay. All these formations, evolution and decay affect the gravitational waves.

3. PROCEDURE

In this section, the tool used to calculate the frequency and energy density results during preheating/reheating for GW will be introduced, then followed by the procedure to approach the goal.

We used the lattice simulation code HLattice written by Huang (2011), which simulates the nonlinear dynamics of the tensor field with a higher accuracy then previous lattice simulation codes such as LATTICEEASY (Felder & Tkachev 2008). It breaks down the space into 3D grid points and evaluate the spacetime curvature change (tensor) and inflaton dynamic (scalar) over time. It uses a sixth-order sympletctic-Runge-Kutta hybrid integrator, and includes the metric perturbation and their feedback (Huang 2011). Redshift was also included in the code, so all result we obtain corresponds to current redshift. The frequency and energy density was calculated using

$$f = \frac{k}{a_j \rho_j^{1/4}} 4 \times 10^{10}\text{Hz}, \quad \Omega_{\text{gw}} h^2 = 9.3 \times 10^{-6} \frac{S_k(\tau_f)}{a_j^4 \rho_j} \quad (5)$$

Where k is the physical wave number, a_j is the scale factor at a moment after radiation dominate, ρ_j is total energy density at a_j, $S_k(\tau_f)$ is transverse-traceless stress-energy tensor. The constant value from the equations comes from the energy density today for radiation and the effectively massless degrees of freedom ratio (Dufaux et al. 2007). A few assumptions were made for preheating/reheating: First, we skip the short matter dominance period, so above equations are referring to radiation dominance. Second, we assumed the equations of state $w = P/\rho$, which the the ratio between pressure and density is 1/3 during this period following radiation dominance value. Third, we assumed the universe at this stage is already flat, so $\Omega_k = 0$ holds. And lastly, conservation of entropy apply in this case (Dufaux et al. 2007).

The main steps to approach our goal are:

- **Understand HLattice:** Reproduce GW signals from Bhoonah et al. (2021)'s paper, understand how frequency and energy density were calculated in equation 5 with all the assumptions.

- **Finding initial conditions:** Calculate initial field value (ϕ_0) with $10^{-6} \leq r \leq 10^{-4}$ to check the in-between process of oscillon formation. ϕ_0 was calculated by maintain a ~ 60 e-folds using mathematica. (See Table 1)

- **Modify HLattice:** Set reduced Planck mass $M_{Pl} = 1024$, grid length $L = 1H^{-1}$, grid resolution $n = 64$, initial momentum $\dot{\phi}_0 = 9.5 \times 10^{-17} M_{Pl}^2$. Adjust the potential model and make the starting condition to $\epsilon \geq 1$.

- **Simulate GW:** Given initial conditions in Tabel 1, run HLattice code to simulate SGWB during preheating/reheating. Summarize and visualize the results in Figure 2 and 3.

4. RESULTS

This section shows the simulated SGWB results obtained with HLattice. The special features and the energy change form oscillon formation will be discussed.

Figure 2 shows the GW signals of T-model for different r values. The x-axis is the frequency in Hz, and the y-axis is the energy density of the signal today. The redshift is calculated within HLattice. The color bar is the scale factor, which is a representation of time. We set $a = 1$ at the end of slow-roll inflation which covers the full oscillation of the inflaton in the potential well. We can see how the signal changes with time as it moves from purple to yellow and converges at yellow. Recall that a smaller r value means a narrower potential and a fainter signal. From the plot, we can see that the initial energy density value is decreasing as r decreases. But we

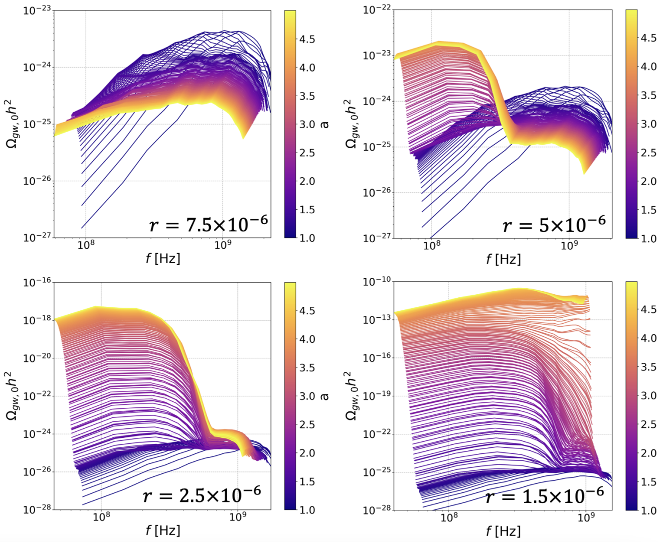

Figure 2. SGWB spectra produced during preheating/reheating after inflation from oscillation of inflaton with a T-model inflationary potential with different r-values. Color bar indicates the evolution of the SGWB as the scale factor involve from 1 to 5 right after the end of slow-roll inflation.

observed a very interesting phenomena: when r reaches about 5×10^{-6}, the energy density has a rapid raise. Figure 2 shows that, a smaller r end up with a higher energy density. Due to the fact that self-resonance dominate the gravitational interaction, the formation of the oscillons result in a dramatic increase of energy density. Once the oscillons settled, GW emissions from individual oscillon will become very low (Lozanov & Amin 2019). Energy density will converge at a specific value as oscillons decay, and that energy value is what would be observe today.

Different inflationary models will have different threshold r-value to allow the formation of oscillon. Specifically for our T-model, r has the upper bound of $\sim 5 \times 10^{-6}$. Because below that value, the field fluctuations will grow rapidly during the post-inflationary phase, less energy goes into gravitational waves, more remains in the scalar field. This will favor the formation of oscillons (Lozanov & Amin 2019), but we are still working on finding numerical connections between the formation of oscillons and r-value in a more general pattern.

We also observed that for $r = 5 \times 10^{-6}$, the signal at smaller frequencies will raise up the energy first, similarly for $r = 2.5 \times 10^{-6}$ case. Lasky et al. (2016) and Lozanov & Takhistov (2023) mentioned

$$\Omega_{gw,0} \sim 10^{-29} \left(\frac{nM}{M_{Pl}} \right)^{7/3} \tag{6}$$

and

$$f_{gw,0} \sim 10^{10} \left(\frac{m^3 M^2}{n M_{Pl}^5} \right)^{1/6} \text{Hz} \tag{7}$$

The common term n, is the number of oscillations it takes for the oscillons to decay. Where $n = m\Delta t/(2\pi)$, depends on mass and lifetime. From these equations, $\Omega_{gw,0} \propto n^{7/3}$, and $f_{gw,0} \propto n^{-1/6}$. Thus, the energy density of oscillon will be smaller at higher frequency. Further address from equation 7, Lozanov & Amin (2019) showed the peak frequency where has the strongest signal is $f_0 \sim \sqrt{\frac{M_{Pl}}{M}} \times 10^8$. With a 1024 M_{Pl} and $M = \sqrt{6\alpha} M_{Pl} = \sqrt{1800 r} M_{Pl}$, the peak frequency is about 5×10^8 Hz. We see from Figure 2 the signals vertically converge at roughly this value.

I have summarized all my simulated signals for T-model with different r-values in Figure 3. The simulated results are bounded by 10^7 Hz to 10^{10} Hz. We see for $r = 10^{-4}$ to $r = 7.5 \times 10^{-5}$, the energy density is bounded between 10^{-21} to about 10^{-25}, and follows the general rule that a smaller r-value will result in a fainter signal (lower energy). The oscillon phenomena occurs for $r \lesssim 5 \times 10^{-6}$, where a sudden raise of energy appears. The signals reach the maximum around $r = 10^{-6}$.

The thick dotted line on the left are the designed detector sensitivities (Moore et al. 2014) of the GW observatories which are LISA (Babak et al. 2021), aLIGO (Abbott et al. 2019), DECIGO, and BBO (Yagi & Seto 2011). Testing processes of these detectors are ongoing, they are also trying to push the limit lower to cover a wider range of energy density. Compare with our simulated result, the frequency range is about 10^{-5} to 10^4 Hz.

The grey horizontal lines represent the SGWB constrains based on current limits from CMB experiments such as Planck satellite (Abbott et al. 2019). Everything above the gray line would be ruled out. With the limitation of 10^{-9}, some of the detectors seems also been ruled out, is it pointless to build them? The colored region is only based on NanoGrav data, which can be bias sometimes. Current researches are also trying to optimize this value by combining COrE and Euclid satellite data with previous CMB and BAO data. More over, recent studies on Atacama Cosmology Telescope (ACT) data also approach the SGWB constrains, data analysis and constrain optimization are in progress (Louis et al. 2025). While the SGWB constrain limits shown are upper bounds, future detectors will be able to measure not only the amplitude but the shape of the signals from a SGWB. Given the link between this shape and

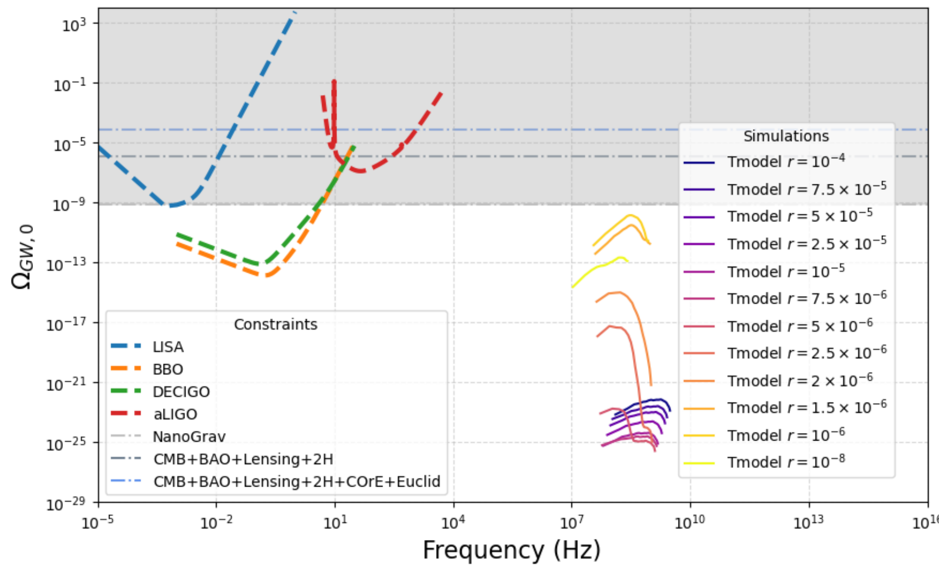

Figure 3. SGWB spectra for oscillating field during preheating/reheating. The final T-model inflationary potential with different r-value was shown on the right side of the plot. Oscillon occurs for $r \lesssim 5 \times 10^{-6}$. The input values and initial conditions for HLattice simulation code is presented in table 1 in Appendix. Dotted horizontal lines are SGWB constrains from Planck satellites and nanoGrav experiments. Dotted curve on the left are current detector sensitivities or designed sensitivities.

our model forecasts, a future detection would be able to distinguish between different models of inflation.

5. CONCLUSION

We used lattice simulations to determine signals during preheating/reheating for T-model inflation and oscillon formation. We see the occurance of oscillons for $r \lesssim 5 \times 10^{-6}$. Our result are in the MHz, GHz range, which matches the estimated range for signals generated during reheating, but too far from the range for current and upcoming gravitational wave observatories.

We can continue our study based on current results by looking at the signals generated during slow-roll inflation. Slow-roll inflation requires different assumptions than the ones mentioned in Section 3: First, we need to consider the affect of GW from the short matter dominance period. This will not affect the high frequency range signals, which is the reason that we can ignore this for preheating/reheating. But it become significant while calculating signals with smaller frequency. Second, $w = -1$ instead of 1/3, because for slow-roll inflation, potential energy is dominant, so $P \propto -V$ and $\rho \propto V$, thus $w = P/\rho \approx -1$ (Ackerman et al. 2011). Third, at the beginning of slow-roll inflation, the curvature might not be 0, we cannot directly conclude that the universe is flat at the beginning of inflation. Equation 5 need to derived using to fit all these new assumptions.

Based on Boyle & Steinhardt (2008)'s paper, the GW signal from slow-roll inflation will have a smaller frequency than preheating/reheating. We expect current and next-generation GW detection instruments such as LIGO (Collaboration et al. 2021), LISA (Braglia et al. 2024) and nanoGrav (Agazie et al. 2023) to help us better understand the inflationary models and inflationary theory by using direct detection methods. These observatories are pushing their limit on the detectable region and trying to cover a wider range of frequency and energy density. It would be very motivated if SGWB signals can be detect in the future, by then, we will be one step closer to the truth of early universe evolution.

ACKNOWLEDGEMENT

I would like to sincerely thank Professor Renée Hložek and Simran Nerval for their invaluable support and guidance during the semester. I am very grateful for their help in research, reading, presentation and academic writing. They made a very positive impact on me and encourage me to continue my journey in astronomy and astrophysics.

REFERENCES

Abbott, B., Abbott, R., Abbott, T., et al. 2016, Physical Review Letters, 116, doi:10.1103/physrevlett.116.061102

—. 2019, Physical Review D, 100, doi:10.1103/physrevd.100.061101

Ackerman, L., Fischler, W., Kundu, S., & Sivanandam, N. 2011, Journal of Cosmology and Astroparticle Physics, 2011, 024–024

Agazie, G., Anumarlapudi, A., Archibald, A. M., et al. 2023, The Astrophysical Journal Letters

Aghanim, N., Akrami, Y., Ashdown, M., et al. 2020, Astronomy amp; Astrophysics, 641, A6

Amin, M. A., Easther, R., Finkel, H., Flauger, R., & Hertzberg, M. P. 2012, Physical Review Letters, 108, doi:10.1103/physrevlett.108.241302

Babak, S., Hewitson, M., & Petiteau, A. 2021, LISA Sensitivity and SNR Calculations, arXiv:2108.01167

Bailes, M., Berger, B. K., Brady, P. R., et al. 2021, Roadmap

Bhoonah, A., Bramante, J., Nerval, S., & Song, N. 2021, ArXiv e-prints, arXiv:2008.12306v2

Boyle, L. A., & Steinhardt, P. J. 2008, Physical Review D, 77, doi:10.1103/physrevd.77.063504

Braglia, M., Calcagnib, G., Franciolinic, G., et al. 2024, ArXiv e-prints, arXiv:2407.04356v1

Collaboration, T. L. S., Collaboration, T. V., & Collaboration, T. K. 2021, ArXiv e-prints, arXiv:2101.12130v1

Dodelson, S., Knox, L., & Kolb, E. W. 1994, Physical Review Letters, 72, 3444–3447

Dufaux, J.-F., Bergman, A., Felder, G., Kofman, L., & Uzan, J.-P. 2007, Physical Review D, 76, doi:10.1103/physrevd.76.123517

Felder, G., & Tkachev, I. 2008, Computer Physics Communications, 178, 929–932

García-Bellido, J., Peloso, M., & Unal, C. 2017, Journal of Cosmology and Astroparticle Physics, 2017, 013–013

Guth, A. H. 1981, Phys. Rev. D, 23, 347

Guth, A. H., & Pi, S.-Y. 1982, Phys. Rev. Lett., 49, 1110

Guzzetti, M. C., Bartolo, N., Liguori, M., & Matarrese, S. 2016, La Rivista del Nuovo Cimento, 39, 399–495

Huang, Z. 2011, ArXiv e-prints, arXiv:1102.0227v2

Lasky, P. D., Mingarelli, C. M., Smith, T. L., et al. 2016, Physical Review X, 6, doi:10.1103/physrevx.6.011035

Louis, T., Posta, A. L., Atkins, Z., et al. 2025, The Atacama Cosmology Telescope: DR6 Power Spectra, Likelihoods and ΛCDM Parameters, arXiv:2503.14452

Lozanov, K. D., & Amin, M. A. 2019, Physical Review D, 99, doi:10.1103/physrevd.99.123504

Lozanov, K. D., & Takhistov, V. 2023, Physical Review Letters, 130, doi:10.1103/physrevlett.130.181002

Moore, C. J., Cole, R. H., & Berry, C. P. L. 2014, Classical and Quantum Gravity, 32, 015014

R. Liddle, A. 1999, ArXiv e-prints, arXiv:9901124v1

Tristram, M., Banday, A., Górski, K., et al. 2022, Physical Review D, 105, doi:10.1103/physrevd.105.083524

Yagi, K., & Seto, N. 2011, Physical Review D, 83, doi:10.1103/physrevd.83.044011

APPENDIX

Table 1. Simulation parameters for T-model inflationary potentials given in eq 3. Where ϕ_0 is the initial field value at the end of slow-roll inflation. The initial kinetic field value is $-9.5 \times 10^{-17} M_{Pl}^2$ for all different α and r. All e-folds values are approximated to 60. Final SGWB signals with these initial conditions are shown in Figure 3

α	r	ϕ_0 (M_{Pl})	E-folds value
3×10^{-2}	10^{-4}	1.82	61.6173
2.25×10^{-2}	7.5×10^{-5}	1.625	60.3534
1.5×10^{-2}	5×10^{-5}	1.385	59.3781
0.75×10^{-2}	2.5×10^{-5}	1.055	60.6234
3×10^{-3}	10^{-5}	0.73	61.8398
2.25×10^{-3}	7.5×10^{-6}	0.647	59.9084
1.5×10^{-3}	5×10^{-6}	0.549	61.7943
0.75×10^{-3}	2.5×10^{-6}	0.41	59.2809
0.6×10^{-3}	2×10^{-6}	0.375	62.425
0.45×10^{-3}	1.5×10^{-6}	0.332	61.8918
3×10^{-4}	10^{-6}	0.28	62.8253
3×10^{-6}	10^{-8}	0.0377	60.9485